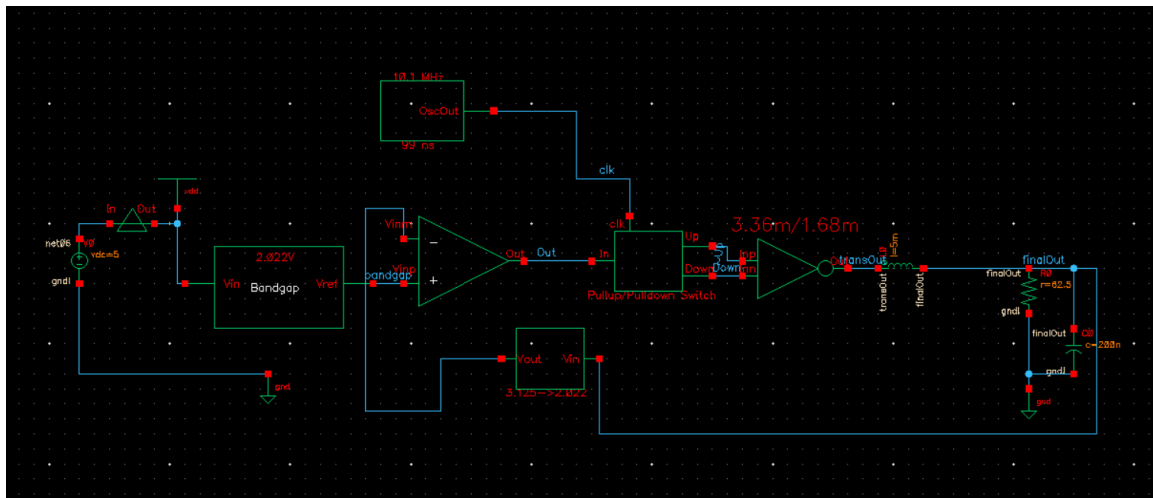


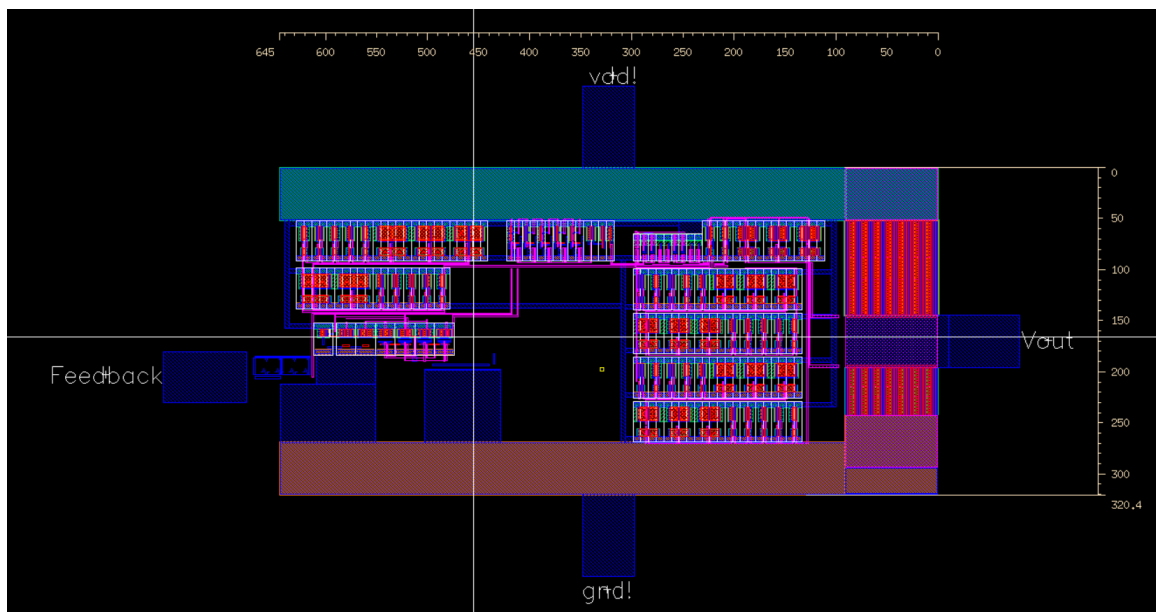
EE421 Final Project: Buck Synchronous SPS

Overview

A buck switching power supply, or buck SPS, steps down an input voltage source and supplies power to a load, such as a laptop or waffle maker. The advantage of a buck SPS over a linear regulator is the low power consumption, and thus high efficiency of the circuit. The only resistive component in the design through which major average current flows is the transistor power switch, which has a calculated resistance of 5Ω , significantly less than the load resistance. This particular design is a synchronous buck, which includes an NMOS power switch for greater efficiency.



Schematic of design, including both on chip and off chip components.



Layout of design, including only on chip components. Dimensions are 645um/320.4um. Primary vdd and gnd wires are 50um wide to avoid electromigration at 50mA. The Vout wire is also 50um wide.

Overall Design Characteristics

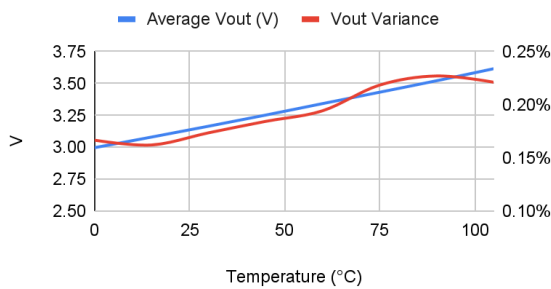
- **Nominal Capabilities* (Spectre)**
 - **Average supplied voltage of 3.124V**
 - Supplied voltage oscillation of $\pm 0.19\%$
 - **Average supplied current of 50 mA**
 - Supplied current oscillation of $\pm 0.19\%$
 - **Minimum inductor current = 49mA (oscillation $\sim 1\text{mA}$)**
 - **Efficiency = 85.25%**
- **General Capabilities** (Ultrasim)**
 - **Average supplied voltage of 2.998-3.616V**
 - Nominal* in Ultrasim is 3.15V
 - **Average supplied current of 10.5-240.4mA**
 - Nominal* in Ultrasim is 50.38 mA
 - **Efficiency $\geq 65.51\%$**
 - Nominal* in Ultrasim is 86.5%
 - **Supplied voltage oscillation of $\leq 0.67\%$**
 - Nominal* in Ultrasim is 0.16%
 - **Average supplied voltage increases linearly with temperature by $0.19\%/^{\circ}\text{C}$ ($0.006\text{V}/^{\circ}\text{C}$)**
- **Maximum/Minimum Average Supplied Current**
 - **Maximum supplied current of 527 mA**
 - Efficiency gradually decreases as R_{load} decreases. Average output voltage suddenly drops at $2.73\Omega < R_{\text{load}} < 5.98\Omega$, so $R_{\text{load minimum}} = 5.98\Omega$
 - Efficiency @ $R_{\text{load minimum}} = 69.3\%$
 - **Minimum supplied current of 4.8 mA**
 - Efficiency gradually decreases as R_{load} increases. Unlike small values of R_{load} , large values of R_{load} have fairly constant average output voltage. $R_{\text{load maximum}} = 653.5\Omega$ was selected on the basis of being the highest tested R_{load} for which the inductor current is always positive, although the supplied voltage oscillation is $\pm 1.64\%$.
 - Efficiency @ $R_{\text{load maximum}} = 46.2\%$
 - The highest R_{load} tested was 3125Ω , which had an average supplied current of 1mA at an efficiency of 15.5%
- **Open Circuit Response ($R_{\text{load}} = 100\text{M}\Omega$)**
 - **Supplied voltage = $3.007\text{V} \pm 20.5\%$**
 - **Supplied current = 0**

*Nominal conditions are 27°C , $R_{\text{load}} = 62.5\Omega$, input voltage 5V

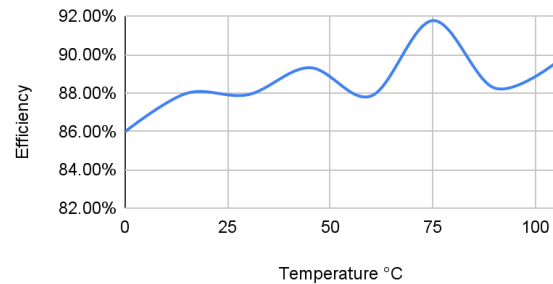
**General conditions are $0\text{--}100^{\circ}\text{C}$, load resistance $13.1\text{--}299\Omega$, input voltage $4\text{--}5.5\text{V}$, all tested independently. The range of load resistances is defined as $(2.1867)R_{\text{load minimum}}$ to $(1/2.1867)R_{\text{load maximum}}$

Graphs of Functionality Under General Conditions**

Supply Voltage Temperature Response

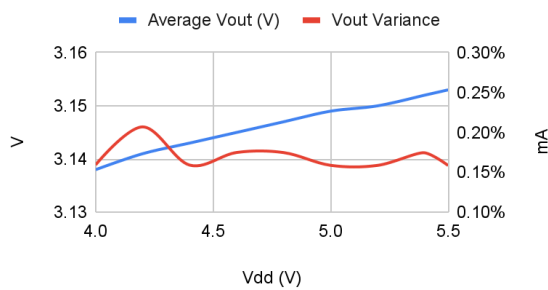


Efficiency Temperature Response

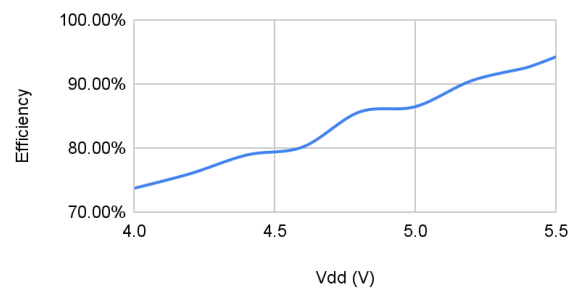


Average supplied voltage and supplied voltage oscillation increase weakly with temperature. Efficiency is not strongly effected.

Supply Voltage Vdd Response

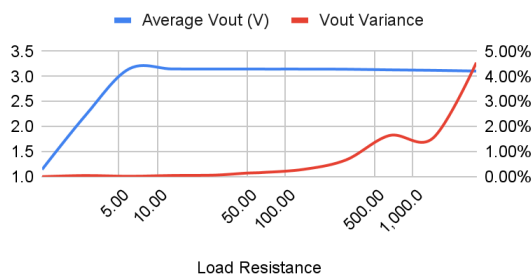


Efficiency Vdd Response

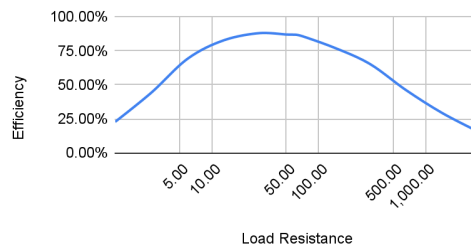


Average supplied voltage increases weakly with increasing Vdd, while the supplied voltage oscillation is invariant. Efficiency increases strongly with increasing Vdd.

Supply Voltage RL Response (log)

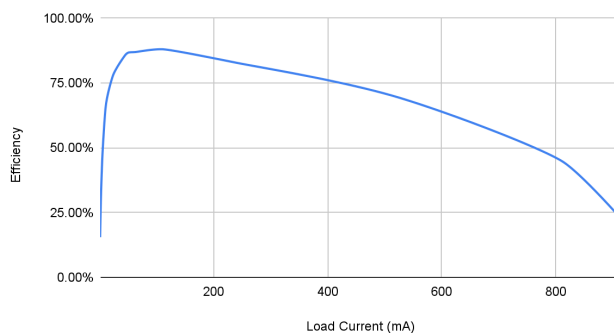


Efficiency RL Response (log)



Average supplied voltage decreases rapidly below $R_{load} \cong 5-10\Omega$, but is fairly stable at higher values. The efficiency peaks at $R_{load} \cong 30\Omega$, then decreases at higher/lower resistances.

Efficiency vs Current



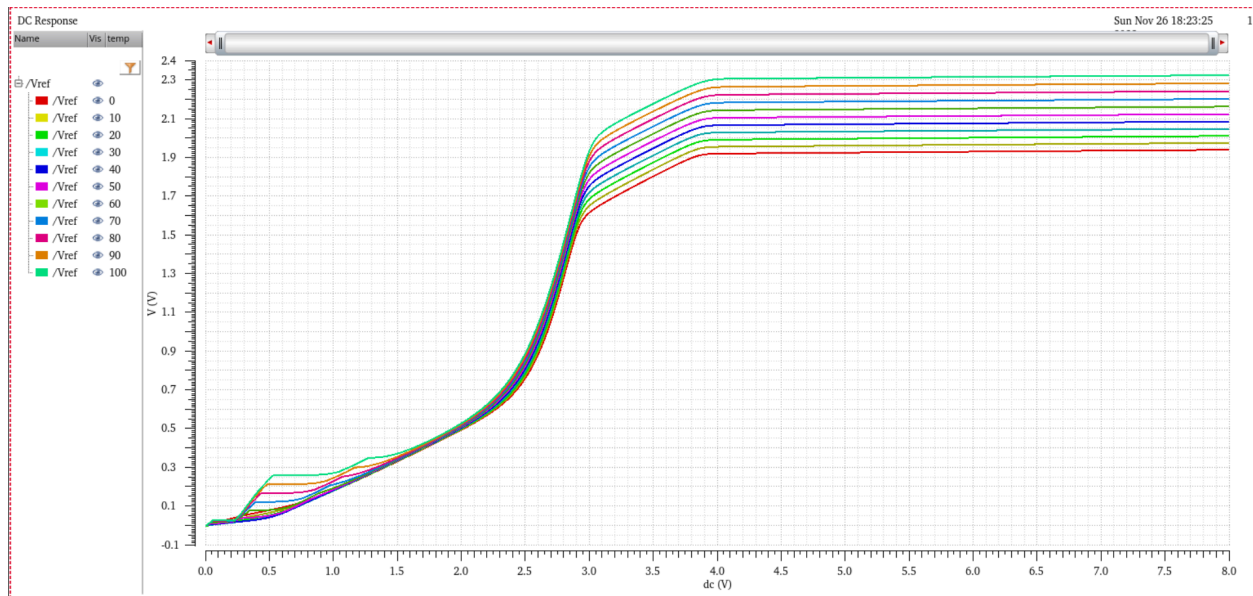
Efficiency decreases close to linearly after its peak, then accelerates as the inductor current goes negative. Despite this trend, the difference in slope isn't substantial enough to say for sure that the negative inductor current is a major cause of efficiency loss.

Individual Component Characteristics and Design Choices

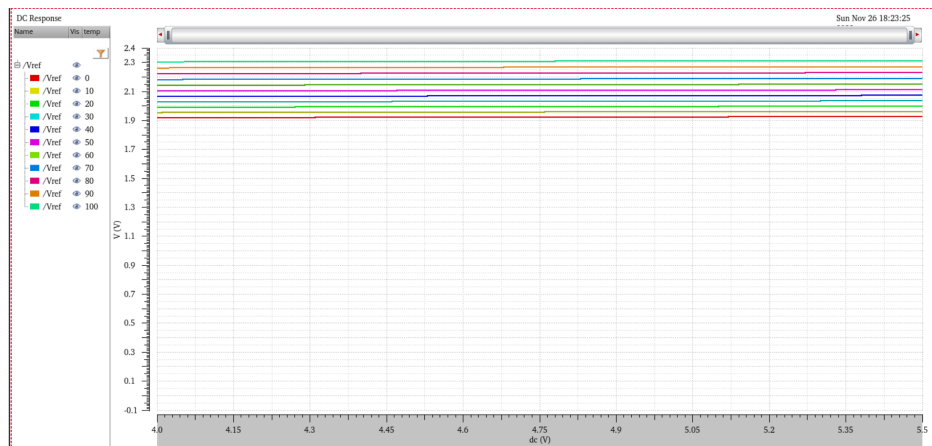
Bandgap

The bandgap voltage reference used in this design outputs $\sim 2.022\text{V}$ for an input voltage of 4-5.5V. It consists of a Thermal Voltage Referenced Self Biasing circuit (PTAT), which causes the reference voltage to increase linearly with temperature ($0.19\%/^{\circ}\text{C}$, or $0.00385\text{V}/^{\circ}\text{C}$). As a result, the total design's average supplied voltage also increases linearly with temperature, at a rate of $0.19\%/^{\circ}\text{C}$ ($0.006\text{V}/^{\circ}\text{C}$).

The diodes in this bandgap are simple 3/0.6 NMOS diodes, and most other parameters were left unchanged from the provided bandgap circuit. However, the change in diode implementation increased both the "turn on" voltage of the reference as well as the nominal* output voltage. The resistor values in both the bandgap and the feedback voltage divider were adjusted slightly to compensate for this change.



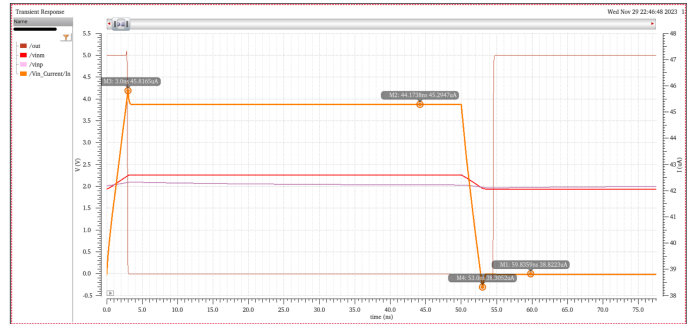
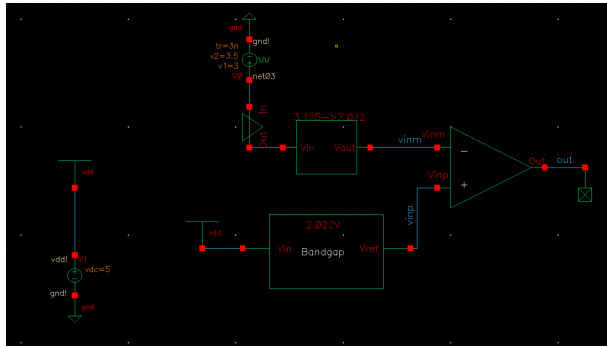
Parametric DC/temperature response of the PTAT based bandgap voltage reference. Temperature ranges from 0-100°C



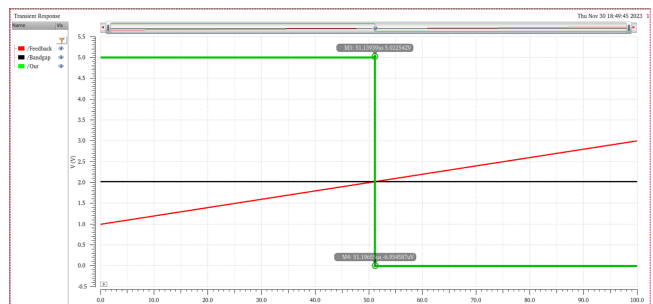
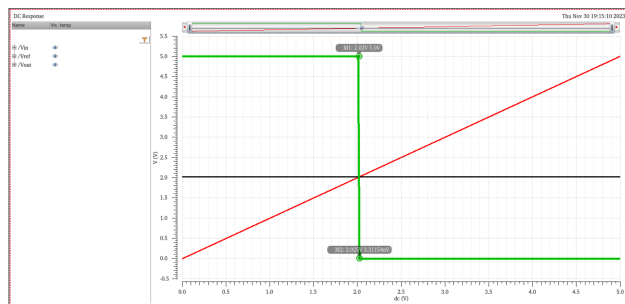
The previous graph scaled to fit the design's general conditions**

Control/Feedback Circuit

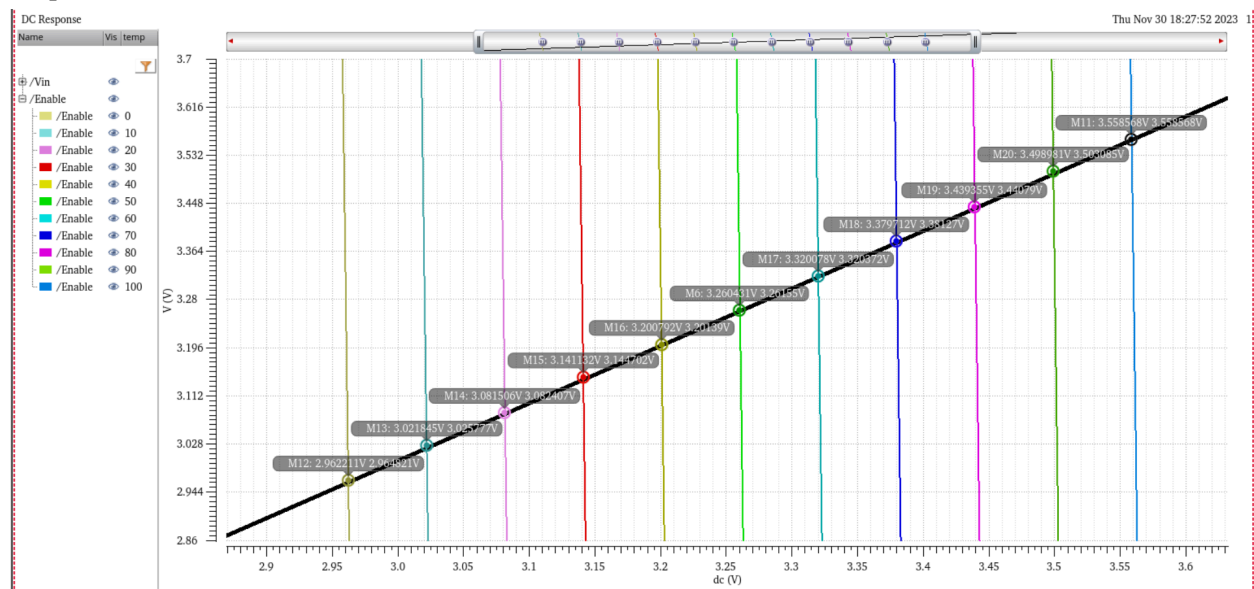
The control feedback circuit specified in part 2 of the project instructions consists of a comparator, voltage divider, and bandgap circuit. With an input varying by $\pm 0.25\text{V}$ (period 100ns, rise/fall time 3ns,) the current drawn by V_{in} varies from 38.3–45.8uA. Smaller voltage variations (the actual design varies by about $\pm 0.006\text{V}$) lead to smaller current variations, without changing the average significantly.



Feedback circuit schematic (left) and current drawn (right)



DC/Temperature (left) and transient (right) comparator response. Notice the comparator is practically temperature invariant.



Temperature variance (DC sweep) of feedback circuit. Switching voltage increases by 0.19%/°C (0.0060V/°C), the same as the SPS' average supplied voltage.

Sizing of power switch transistors

The power switch consists of a 3360um PMOS and 1680um NMOS (length=600nm.) The approximate resistance of each MOSFET is 5Ω . Lower MOSFET resistance means less power is dissipated by the switch, which means higher efficiency. This is particularly noticeable at small load resistances.

Theoretically, using power MOSFETs of such large widths necessitates large driving inverters on the right side of the pullup/pulldown circuit. A test was done with an additional pair of 420u/210u inverters added after each of the 96u/48u inverters currently present in the design, in an effort to better drive the power MOSFETs. Overall efficiency decreased and there was little noticeable benefit. As a result, only the 96u/48u driving inverters were kept.

Naturally, when an relatively small inverter drives a MOSFET (or capacitor) of large size, the delay can be substantial. The rise and fall times of V_{ov} , or the time to turn on and off, for both the PMOS and NMOS in the power switch range from ~ 2 -3ns, relatively small compared to the 100ns of hysteresis. A 30ns delay is implemented in the pullup/pulldown circuit to avoid contention current, which is admittedly a bit excessive. The main cost of such a large delay is the physical size of the buffers, which make the pullup/pulldown circuit the largest component in the design by area.

Sizing of off chip L,C, and selection of switching frequency

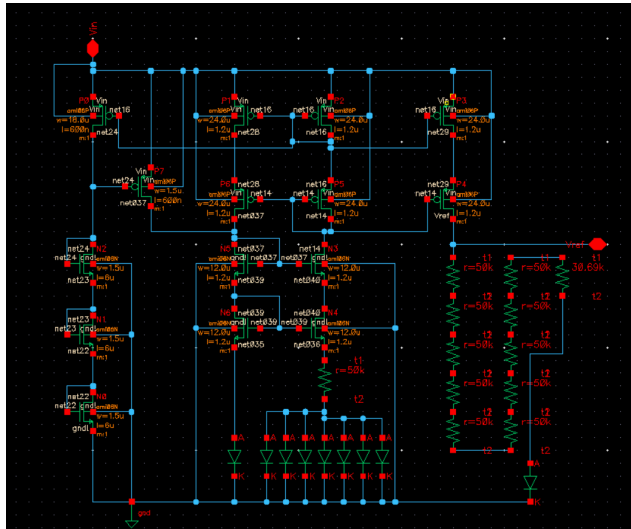
Initially, the inductance and capacitance values of the off chip output circuit were selected mathematically. According to the calculations, $L=1.16\mu\text{H}$ for an inductor current oscillation of $\pm 50\text{mA}$, and $L=96.7\mu\text{H}$ for an inductor current oscillation of $\pm 1\text{mA}$. $C=4.75\text{nF}$ for a V_{out} oscillation of $\pm 0.1\%$, given $L=96.7\mu\text{H}$. Unfortunately, both these values ended up being far too low, so the final values used were selected through trial and error.

A larger inductor results in less inductor current oscillation, which helps prevent current flowing backwards through the power switch, which would decrease efficiency. $L=5\text{mH}$ corresponds with a nominal* inductor current of 48.97-51.06mA, or very nearly $\pm 1\text{mA}$. In general conditions**, this inductor current oscillation is roughly proportional to average supplied voltage and is mostly independent from capacitor size.

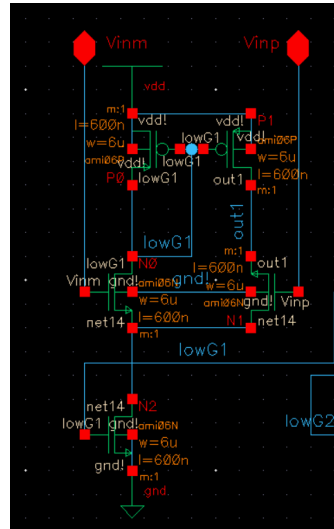
A larger capacitor results in smaller supplied voltage oscillation, though testing revealed rapidly diminishing returns after about 200nF. Increasing the capacitance from 200nF to 1600nF resulted in a V_{out} oscillation reduction of only 25%. The final design ended up using a filter capacitance of 200nF, as it's about the largest the filter capacitor can be without excessive simulation times.

Higher switching frequency also results in greatly decreased voltage oscillation. However, higher frequency leads to greater power consumption and increased risk of timing issues. A switching frequency of 10.1MHz was selected to compliment the 200nF filter capacitor, which results in a nominal* V_{out} oscillation of $\pm 0.19\%$, while most of the power loss still occurs in the power switch.

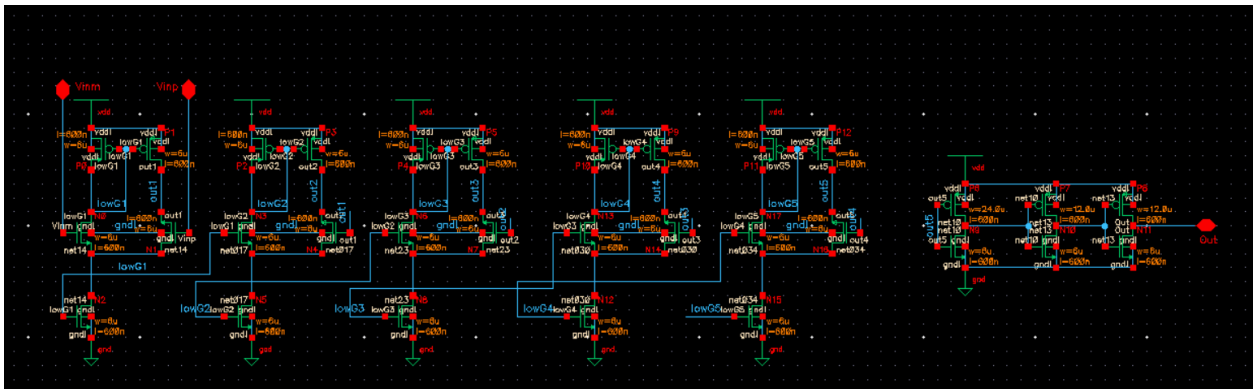
Schematics



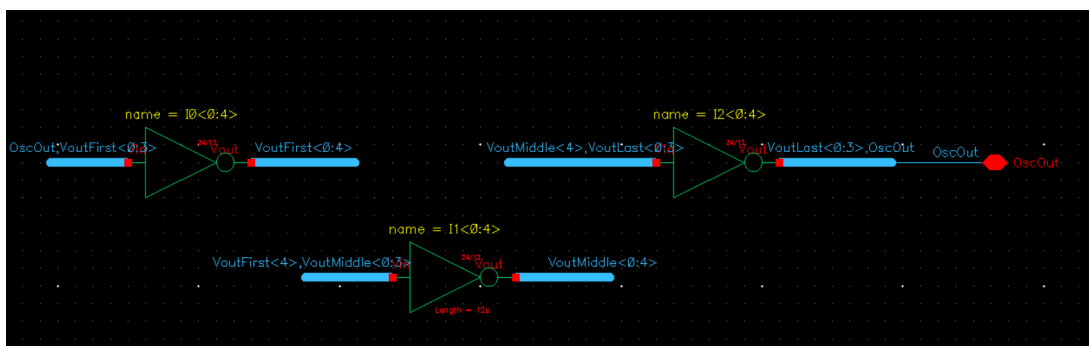
Bandgap



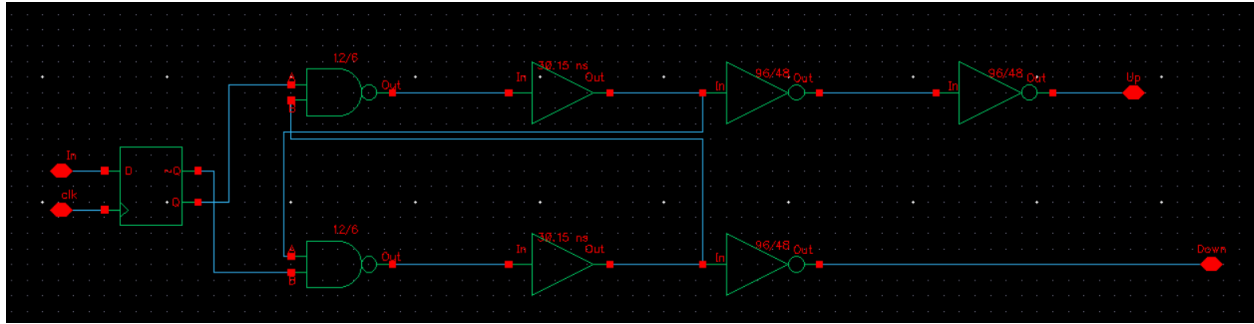
Self-biased diff-amp



Comparator, consisting of 5 self biased diff-amps and 3 inverters

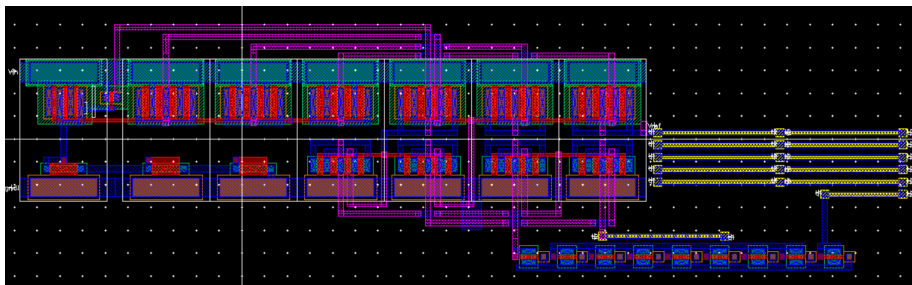


5-5-5 ring oscillator (L= 0.6u, 12u, 0.6u) (10.1 MHz)

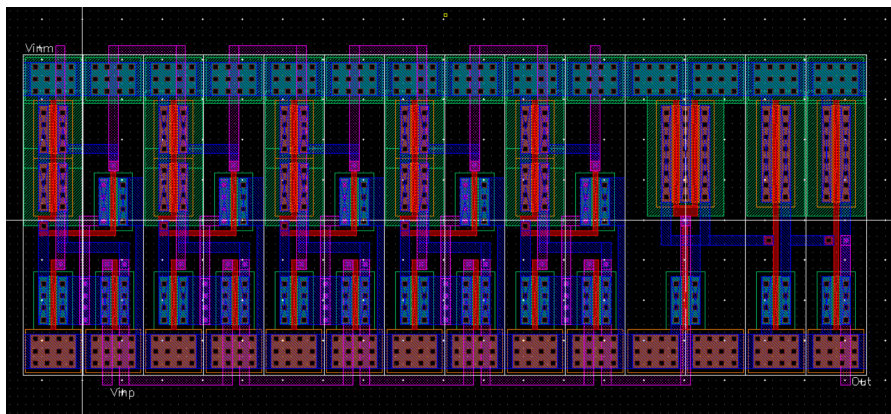


Pullup/Pulldown circuit. Clocked DFF allows 100ns hysteresis. 30ns buffers to remove contention current

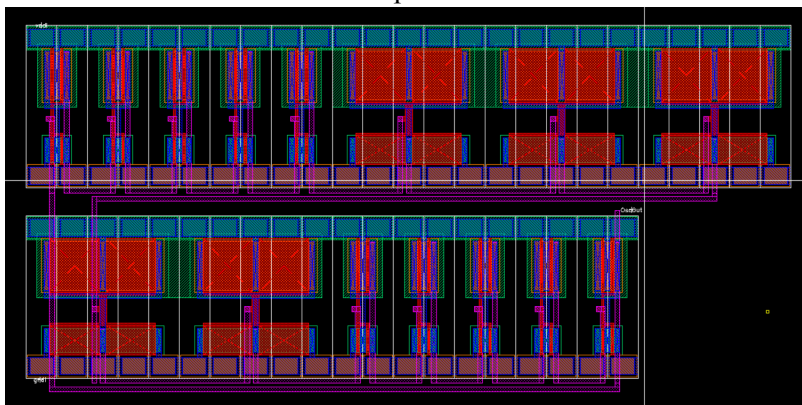
Layouts



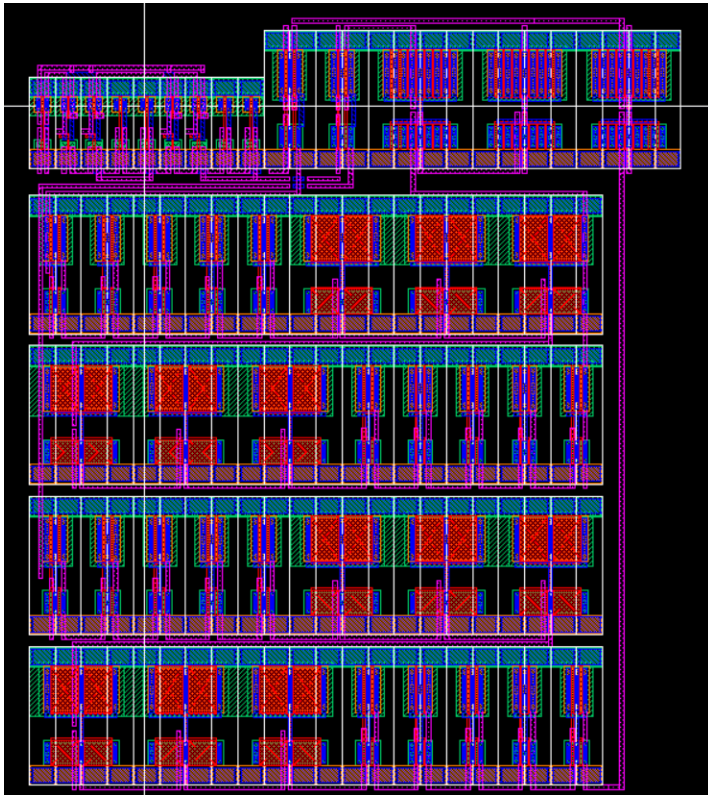
Bandgap



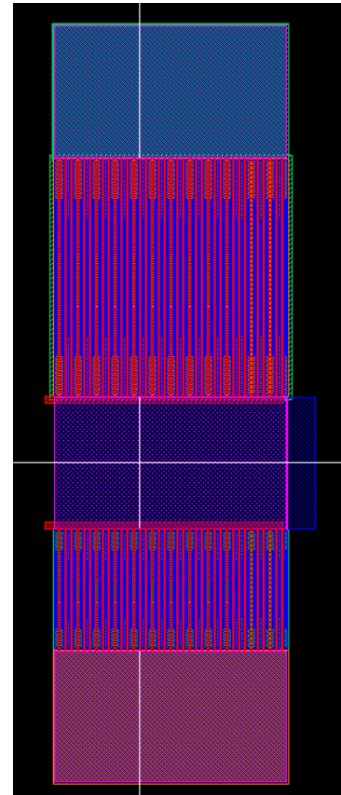
Comparator



Ring Oscillator



Pullup/Pulldown. The bottom 4 layers are a pair of 30ns buffers.



Power Switch Transistors
($W_p=90.75u$, $m=37$)

Conclusion

While this buck SPS design is certainly not perfect, it is functional enough, with an efficiency comparable to that of some commercial buck converters under nominal conditions. The design doesn't work well in suboptimal conditions, particularly in terms of temperature variance. The best way to decrease the temperature variance of the design would be to include a CTAT circuit in the bandgap voltage reference.

While the lack of adjustable output voltage and somewhat narrow range of viable output currents mean this design can't power most devices, there are certainly some it would work just fine for. For example, AL1411A-WNL-020MA, an LED manufactured by Solidlite and available on DigiKey, operates nominally at 3.15V and 20mA. At 3.125V and 20.16mA, this buck SPS would be able to power the LED with about 85% efficiency.